MLP (ov4) | 96p96c4g | numa | collective vs p2p1 | rep 1 | call 6 / MAIA step 189

Input + ACK
Credit wait
Output
Host inference
H2D transfer
Torch forward
D2H transfer
Credit send

WG 0 Controller r96

r185 [Worker]
r184 [Worker]
r177 [Worker]
r176 [Worker]
r169 [Worker]
r168 [Worker]
r161 [Worker]
r160 [Worker]
r153 [Worker]
r152 [Worker]
r145 [Worker]
r144 [Worker]
r137 [Worker]
r136 [Worker]
r129 [Worker]
r128 [Worker]
r121 [Worker]
r120 [Worker]
r113 [Worker]
r112 [Worker]
r105 [Worker]
r104 [Worker]
r97 [Worker]
r96 [GPU Ctrl]

Collective gather end
Collective inference end
Collective scatter end

r92 [Worker]
r88 [Worker]
r84 [Worker]
r80 [Worker]
r76 [Worker]
r72 [Worker]
r68 [Worker]
r64 [Worker]
r60 [Worker]
r56 [Worker]
r52 [Worker]
r48 [Worker]
r44 [Worker]
r40 [Worker]
r36 [Worker]
r32 [Worker]
r28 [Worker]
r24 [Worker]
r20 [Worker]
r16 [Worker]
r12 [Worker]
r8 [Worker]
r4 [Worker]
r0 [Worker]

WG 1 Controller r98

r187 [Worker]
r186 [Worker]
r179 [Worker]
r178 [Worker]
r171 [Worker]
r170 [Worker]
r163 [Worker]
r162 [Worker]
r155 [Worker]
r154 [Worker]
r147 [Worker]
r146 [Worker]
r139 [Worker]
r138 [Worker]
r131 [Worker]
r130 [Worker]
r123 [Worker]
r122 [Worker]
r115 [Worker]
r114 [Worker]
r107 [Worker]
r106 [Worker]
r99 [Worker]
r98 [GPU Ctrl]

Collective gather end
Collective inference end
Collective scatter end

r93 [Worker]
r89 [Worker]
r85 [Worker]
r81 [Worker]
r77 [Worker]
r73 [Worker]
r69 [Worker]
r65 [Worker]
r61 [Worker]
r57 [Worker]
r53 [Worker]
r49 [Worker]
r45 [Worker]
r41 [Worker]
r37 [Worker]
r33 [Worker]
r29 [Worker]
r25 [Worker]
r21 [Worker]
r17 [Worker]
r13 [Worker]
r9 [Worker]
r5 [Worker]
r1 [Worker]

WG 2 Controller r100

r189 [Worker]
r188 [Worker]
r181 [Worker]
r180 [Worker]
r173 [Worker]
r172 [Worker]
r165 [Worker]
r164 [Worker]
r157 [Worker]
r156 [Worker]
r149 [Worker]
r148 [Worker]
r141 [Worker]
r140 [Worker]
r133 [Worker]
r132 [Worker]
r125 [Worker]
r124 [Worker]
r117 [Worker]
r116 [Worker]
r109 [Worker]
r108 [Worker]
r101 [Worker]
r100 [GPU Ctrl]

Collective gather end
Collective inference end
Collective scatter end

r94 [Worker]
r90 [Worker]
r86 [Worker]
r82 [Worker]
r78 [Worker]
r74 [Worker]
r70 [Worker]
r66 [Worker]
r62 [Worker]
r58 [Worker]
r54 [Worker]
r50 [Worker]
r46 [Worker]
r42 [Worker]
r38 [Worker]
r34 [Worker]
r30 [Worker]
r26 [Worker]
r22 [Worker]
r18 [Worker]
r14 [Worker]
r10 [Worker]
r6 [Worker]
r2 [Worker]

WG 3 Controller r102

r191 [Worker]
r190 [Worker]
r183 [Worker]
r182 [Worker]
r175 [Worker]
r174 [Worker]
r167 [Worker]
r166 [Worker]
r159 [Worker]
r158 [Worker]
r151 [Worker]
r150 [Worker]
r143 [Worker]
r142 [Worker]
r135 [Worker]
r134 [Worker]
r127 [Worker]
r126 [Worker]
r119 [Worker]
r118 [Worker]
r111 [Worker]
r110 [Worker]
r103 [Worker]
r102 [GPU Ctrl]

Collective gather end
Collective inference end
Collective scatter end

r95 [Worker]
r91 [Worker]
r87 [Worker]
r83 [Worker]
r79 [Worker]
r75 [Worker]
r71 [Worker]
r67 [Worker]
r63 [Worker]
r59 [Worker]
r55 [Worker]
r51 [Worker]
r47 [Worker]
r43 [Worker]
r39 [Worker]
r35 [Worker]
r31 [Worker]
r27 [Worker]
r23 [Worker]
r19 [Worker]
r15 [Worker]
r11 [Worker]
r7 [Worker]
r3 [Worker]

0
40
80
120
160
200

ms from each WG's own controller ml_step_start (independent zero; estimated alignment)